

NOTES IN ANALYSIS I

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Abstract. These notes follow the Real Analysis course by Dr. Emanuel Carneiro - ICTP. All errors are my own.

1 Basic concepts

Think about describing the Borel σ -algebra.

2 Measure

2.1 Remarks

- $A \subseteq B \implies \mu(A) \leq \mu(B)$
 $B = (B \setminus A) \cup A$
 $\mu(B) = \mu(B \setminus A) + \mu(A)$ Observe $(\mu(B \setminus A) \geq 0)$, and can even be infinite
- If $\mu(\Omega) < \infty$, this is a finite measure. (If $\mu(\Omega) = 1$, a probability measure.)
- If $\Omega = \bigcup_{i=1}^{\infty} A_n$, $\mu(A_n) < \infty$, μ is σ -finite.

2.2 Dirac measure

tbd and proved